

FPGA-Based Virtual Quadrature Encoder Synthesis from Hall Effect Sensors

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Abstract— This paper presents the design of a virtual incremental encoder based on three-phase Hall-effect sensor feedback. The proposed system converts the six valid Hall sensor states of a brushless DC motor into two-phase quadrature encoder signals, commonly referred to as Channel A and Channel B. The system is designed for implementation on a field-programmable gate array using the VHDL hardware description language. The proposed architecture includes Hall-input synchronization, Hall-state transition detection, rotational direction identification, time-interval measurement, pulse-period calculation, and quadrature signal generation. A two-stage synchronization technique is used to reduce metastability caused by asynchronous Hall inputs. The detected Hall transitions are used to determine the direction of rotation and estimate motor speed. Based on this information, a quadrature-state generator produces the corresponding incremental encoder signals. The proposed system provides a low-cost alternative to a dedicated incremental encoder for motor-control and embedded motion-control applications.

Keywords— Hall sensor, virtual encoder, incremental encoder, quadrature signal, FPGA, VHDL, brushless DC motor, motor control.

I. INTRODUCTION

A virtual encoder provides an alternative approach by digitally converting the Hall sensor signals into quadrature encoder outputs. The proposed design receives the three Hall signals, synchronizes them with the FPGA clock, determines the direction of rotation from the Hall-state sequence, measures the time between consecutive Hall transitions, and generates corresponding Channel A and Channel B signals. The design is implemented in VHDL and is intended for FPGA-based motor-control applications.

The main objective of this work is to develop a simple and synthesizable digital architecture that can generate speed-dependent quadrature encoder signals without requiring an additional physical

encoder. The proposed design is particularly suitable for applications where coarse rotor-position information is sufficient and where Hall sensors are already available.

II. SYSTEM DESCRIPTION

The proposed virtual encoder consists of several functional stages. The first stage synchronizes the three asynchronous Hall sensor inputs with the FPGA system clock. The second stage stores consecutive Hall states and detects changes between them. A transition decoder then identifies the direction of rotation by comparing the detected sequence with predefined clockwise and counter-clockwise Hall sequences.

The time interval between consecutive Hall transitions is measured using a digital counter. This interval is related to motor speed because Hall transitions occur at a rate proportional to rotor speed. The measured interval is scaled to obtain a pulse period for the quadrature generator. Finally, a two-bit quadrature state machine generates the sequence required for Channel A and Channel B.

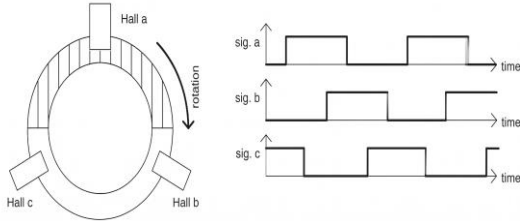
The complete process is performed synchronously using a 25 MHz clock. The clock period is therefore 40 ns. This clock provides the time reference for Hall-state sampling, interval measurement, and quadrature signal generation.

III. HALL SENSOR STATE DETECTION

A three-phase Hall sensor interface produces a three-bit digital value. For a conventional three-phase brushless DC motor, six Hall states are normally considered valid. The clockwise sequence used in the proposed design is 001, 011, 010, 110,

100, 101, and then 001. The reverse sequence is 001, 101, 100, 110, 010, 011, and then 001.

The Hall states 000 and 111 are generally considered invalid in a standard three-sensor configuration. These states may indicate a sensor fault, wiring problem, startup condition, or transient disturbance. The proposed direction detector compares each new Hall state with the preceding state. If the transition belongs to the predefined clockwise sequence, the direction is assigned as clockwise. If it belongs to the counter-clockwise sequence, the direction is assigned as counter-clockwise.



This method provides a simple method of determining direction without requiring a separate direction sensor. Since the decoder considers the order of Hall transitions rather than only the current state, it can distinguish forward and reverse rotation.

IV. HALL INPUT SYNCHRONIZATION

The Hall sensor outputs are asynchronous relative to the FPGA system clock. Sampling an asynchronous signal directly with synchronous logic can cause metastability. Metastability may result in unpredictable behavior or incorrect state detection.

To reduce this possibility, the proposed architecture uses two sequential synchronization stages. The first stage samples the external Hall signals, and the second stage samples the output of the first stage. The second-stage output is then used by the Hall-state detection logic. This technique provides a stable representation of the Hall inputs in the FPGA clock domain.

The two-stage synchronizer does not eliminate metastability completely, but it significantly reduces the probability that metastability will propagate to the remainder of the design. Additional digital filtering may be required in applications where the Hall signals contain significant noise or contact-related disturbances.

V. DIRECTION DETECTION

Direction detection is performed by comparing the current synchronized Hall state with the previous synchronized Hall state. A change between these states indicates a possible Hall transition. The transition is then compared with the valid transition tables.

For clockwise rotation, the Hall states follow the sequence 001 to 011, 011 to 010, 010 to 110, 110 to 100, 100 to 101, and 101 to 001. For counter-clockwise rotation, the sequence occurs in the opposite order. When a valid clockwise transition is detected, the direction signal is set to clockwise. When a valid counter-clockwise transition is detected, the direction signal is set to counter-clockwise.

If the transition does not match either valid sequence, the direction is not changed. This behaviour prevents an invalid Hall transition from causing an unintended reversal of the virtual encoder output. However, for a more robust implementation, invalid transitions should also be detected and reported through a fault signal or used to disable the encoder output.

VI. SPEED MEASUREMENT

The speed estimation process is based on measuring the time interval between successive Hall transitions. Counter increments on every rising edge of the 25 MHz system clock. When a Hall transition occurs, the counter value represents the number of clock cycles since the previous transition.

The time between Hall transitions can be expressed as the product of the counter value and the system-clock period. Since the clock period is 40 ns, a counter value of one corresponds to 40 ns, while a counter value of 2500 corresponds to 100 μ s.

After a valid Hall transition is detected, the measurement counter is reset and begins counting the interval to the next transition. The measured interval is used to estimate the motor speed. A shorter interval indicates a higher motor speed, while a longer interval indicates a lower motor speed.

The design also includes a timeout mechanism. If no Hall transition occurs before the timeout limit is reached, the system assumes that the motor has stopped or that the Hall signal has been lost. In this

condition, the generated pulse period is cleared and quadrature signal generation is suspended.

It should be noted that the value 2500 clock cycles corresponds to 100 μ s at a 25 MHz clock frequency. Therefore, if a 10ms timeout is required, the counter threshold must be set to 250,000 clock cycles.

VII. PULSE-PERIOD CALCULATION

The measured Hall-transition interval is converted into the period used by the quadrature signal generator. The proposed design applies a fixed-point scaling operation to the measured timer value. The scaling coefficient is implemented using integer multiplication followed by a binary right shift.

This method avoids the use of floating-point arithmetic and is therefore suitable for FPGA implementation. The scaling coefficient determines the relationship between the Hall-transition frequency and the generated quadrature-transition frequency. The coefficient must be selected according to the motor pole-pair number, the number of Hall transitions per revolution, the required encoder resolution, and the type of quadrature decoding used by the receiving system. The pulse period is represented in clock cycles. Consequently, the generated quadrature frequency is quantized according to the FPGA clock frequency. At high motor speeds, the calculated period may become small, leading to quantization error. At low speeds, the large counter range provides adequate measurement resolution.

VIII. QUADRATURE SIGNAL GENERATION

The quadrature generator uses four output states. These states represent the phase relationship between Channel A and Channel B. The output sequence follows a Gray-code pattern in which only one channel changes between adjacent states.

The four quadrature states are 00, 01, 11, and 10, depending on the selected channel-mapping convention. During clockwise rotation, the internal state advances through the sequence in one direction. During counter-clockwise rotation, the state advances in the reverse direction.

A generation counter determines when the next quadrature state transition should occur. When the generation counter reaches the calculated pulse period, it is reset and the quadrature state is advanced or decremented according to the detected direction. If the motor stops or the pulse period becomes zero, the generation counter is reset and the quadrature state remains unchanged.

The generated signals therefore provide both direction and speed information to an external incremental-encoder interface.

IX. TIMEOUT AND STOP DETECTION

The virtual encoder must stop producing pulses when the motor stops. Otherwise, the last measured speed would continue to be reproduced indefinitely. To avoid this condition, the design includes a timeout counter.

The timeout counter increments while no Hall transition is detected. If the counter reaches the selected threshold, the pulse period is set to zero and the quadrature generator is disabled. When a new valid Hall transition occurs, the timer is reset and normal operation resumes.

The timeout threshold should be selected according to the minimum expected motor speed. If the threshold is too short, the encoder may stop during normal low-speed operation. If the threshold is too long, the system may continue producing pulses for an undesirably long period after the motor has stopped.

X. VERIFICATION METHODOLOGY

The proposed design can be verified through VHDL simulation and FPGA hardware testing. The simulation should initially verify the reset condition. During reset, all internal registers should be initialized, the pulse period should be zero, and both quadrature outputs should be low.

The clockwise Hall sequence should then be applied with controlled time intervals. The simulation should verify that the direction detector identifies clockwise rotation and that the quadrature outputs change according to the expected phase sequence. The counter-clockwise sequence should be applied in a separate test to verify reverse operation.

Speed variation can be tested by changing the time interval between Hall transitions. When the Hall transitions occur more frequently, the quadrature output should also increase in frequency. When the transitions occur less frequently, the generated quadrature frequency should decrease.

The timeout function should be tested by holding the Hall input at a constant valid state for longer than the selected timeout. The expected result is that the quadrature outputs stop changing. Invalid states such as 000 and 111 should also be tested to determine whether the design behaves as intended under abnormal input conditions.

Hardware verification should include measurement of the Hall input timing, quadrature output frequency, output duty cycle, direction response, and stop response using a logic analyzer or oscilloscope.

XI. LIMITATIONS

The proposed virtual encoder has several limitations. First, the achievable position accuracy is limited by the resolution of the Hall sensors. Three Hall sensors provide six valid transitions per electrical revolution, which is significantly lower than the resolution of a dedicated optical or magnetic incremental encoder.

Second, the intermediate quadrature transitions are generated through timing interpolation. They do not represent directly measured mechanical position. Any variation in motor speed between Hall transitions may therefore produce position error.

Third, the pulse-period calculation uses integer arithmetic. This introduces quantization, particularly at high speeds when the measured period becomes small. The output frequency may also exhibit timing variation if the motor speed changes significantly between Hall transitions.

Fourth, the initial Hall state after reset may not represent a valid transition. A startup routine should acquire a valid Hall state before enabling direction detection and speed measurement.

Finally, invalid Hall transitions are not fully isolated in the basic architecture. A production design should include explicit validation of the current and previous states and should provide a fault indication for illegal Hall combinations.

XII. RECOMMENDED IMPROVEMENTS

The design can be improved by adding explicit validation for all Hall states and allowing only valid adjacent transitions to update the speed and direction logic. A start up state-acquisition circuit can be used to wait for the first valid Hall state before beginning normal operation.

The timeout value and pulse-period scaling factor should be implemented as configurable parameters. This would allow the same architecture to be adapted to different motor types, clock frequencies, and encoder resolutions.

A digital noise filter can also be added to prevent short glitches from being interpreted as Hall transitions. In addition, the design may include an error output that identifies invalid Hall states, missing transitions, or unexpected state changes.

For systems requiring improved speed accuracy, a more advanced fixed-point calculation or a reciprocal-period measurement method may be used. Hardware testing should also be performed to determine the actual output jitter, maximum operating speed, minimum operating speed, and response time during acceleration and deceleration.

XIII. CONCLUSION

A VHDL-based virtual incremental encoder using three-phase Hall sensor feedback has been presented. The design converts the six-state Hall sequence of a motor into two-phase quadrature encoder signals. It uses synchronized Hall inputs, transition-based direction detection, time-interval measurement, fixed-point pulse-period calculation, and Gray-code quadrature generation.

The proposed architecture provides a practical and low-cost solution for applications that require incremental encoder feedback but already contain Hall-effect sensors. It can be implemented on an FPGA with relatively simple digital logic and can be adapted to different motor-control requirements through changes to the timeout value, scaling coefficient, and output-generation parameters.

Although the design cannot provide the same absolute resolution as a dedicated encoder, it can provide useful speed and direction feedback for many embedded motor-control applications. Future work should focus on experimental hardware

validation, improved Hall-state filtering, configurable interpolation, invalid-state detection, and evaluation of the position error under variable-speed operating conditions.

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